

Capstone Analysis Queries

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Bring in Cleaned & Transformed Dataset

```
Dataset <- read.csv("Cleaned_Transformed_Dataset_Klepps.csv")
```

Import Needed Libraries

```
library(dplyr)
library(tidyr)
library(ggplot2)
```

Exploratory Analysis

Counts

```
Dataset |>
  summarize(unique_count=n_distinct(author))
```

```
##   unique_count
## 1           762
```

```
Dataset |>
  summarize(unique_count=n_distinct(title))
```

```
##   unique_count
## 1          1110
```

```
Dataset |>
  summarize(unique_count=n_distinct(genre))
```

```
##   unique_count
## 1             31
```

Authors

How many times each Author appears (output hidden due to hundreds of lines)

```
author_counts <- Dataset |>
  count(author, name = "count")

author_counts
```

Most popular Authors (output hidden due to hundreds of lines)

```
author_rank <- Dataset |>
  count(author, name = "count") |>
  arrange(desc(count))

author_rank
```

Titles

How many times each Title appears

slightly redundant to above query - verifies each title only appears once?

```
title_counts <- Dataset |>
  count(title, name = "count")
```

Title Ratings

```
title_ratings <- Dataset |>
  arrange(desc(rating)) |>
  mutate(order_rating = row_number())

result_ratings <- title_ratings |>
  select(title, author, rating)
```

Genres

How many time each Genre appears

```
genre_counts <- Dataset |>
  count(genre, name = "count") |>
  arrange(desc(count))
```

How many time each Subgenre appears

```
subgenre_counts <- Dataset |>
  count(subgenre, name = "count") |>
  arrange(desc(count))
```

Genres & associated Subgenres

```
genres_subgenres_nested <- Dataset |>
  group_by(genre) |>
  summarize(subgenres = list(unique(subgenre)))
```

```
split(unique(Dataset$subgenre), Dataset$genre)
```

Refined Analysis & Querying Setup

Packages

```
library(tidyverse)
library(scales)
library(stringr)
```

Theme Creation

```
theme_sage_purple <- function() {
  theme_bw() +
  theme(
    # Axis Text
    axis.text.y = element_text(size = 9, lineheight = 1.1),
    axis.text.x = element_text(size = 10),

    # Titles
    plot.title = element_text(size = 16, face = "bold", hjust = 0.5, margin = margin(b = 10)),
    axis.title.y = element_text(size = 12, face = "bold"),
    axis.title.x = element_text(size = 12, face = "bold"),

    # Background
    panel.background = element_rect(fill = "#DDE8D8", color = NA),
    plot.background = element_rect(fill = "#DDE8D8", color = NA),

    # Gridlines
    panel.grid.major = element_line(color = "white"),
    panel.grid.minor = element_line(color = "white"),

    #Margins
```

```

    plot.margin = margin(10, 40, 10, 10)
  )
}

```

Refined Queries

1. Which genres generate the most total revenue? (using number of reviews multiplied by final selling price)

```

Dataset <- Dataset |>
  mutate(revenue = final_price * reviews_count)

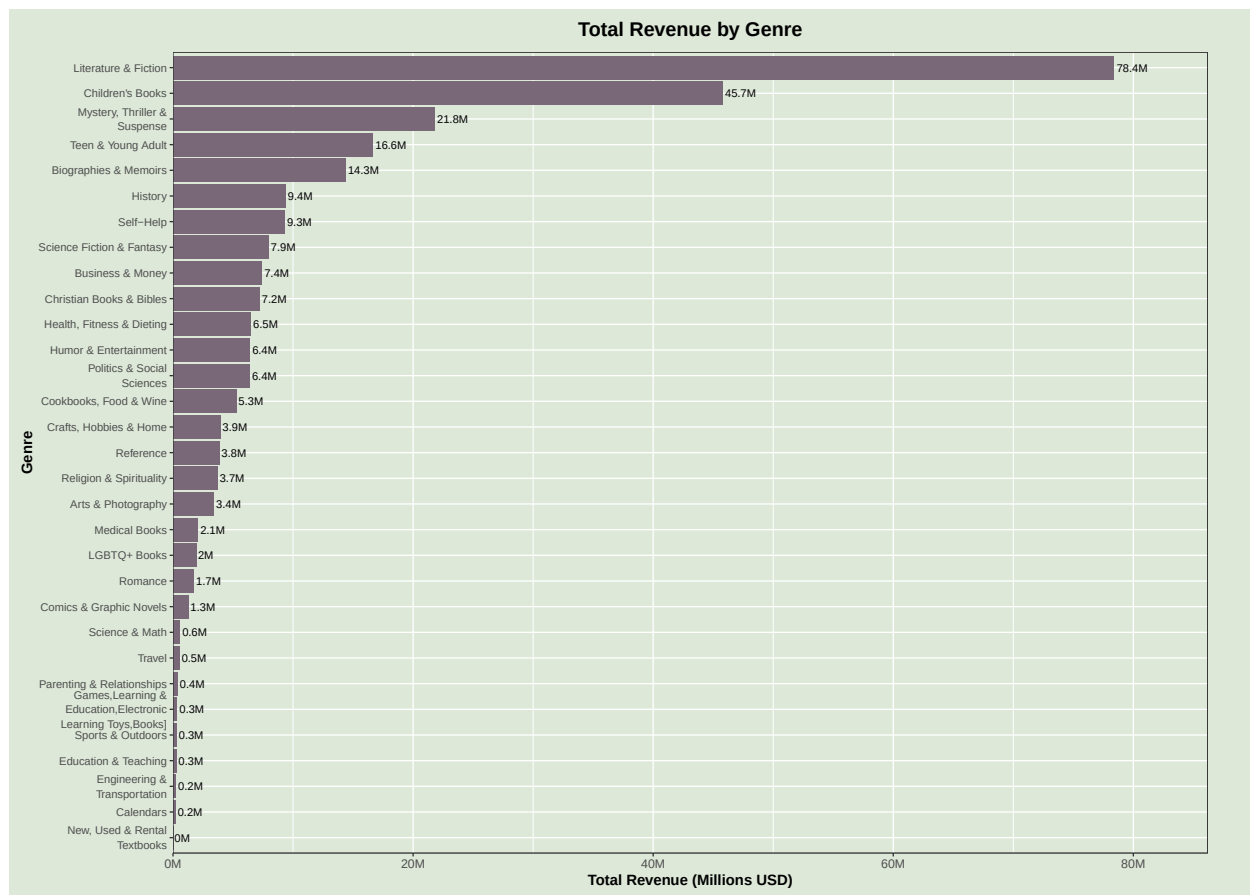
genre_revenue <- Dataset |>
  group_by(genre) |>
  summarize(total_revenue = sum(revenue, na.rm = TRUE)) |>
  arrange(desc(total_revenue))

```

```

ggplot(genre_revenue, aes(x = reorder(str_wrap(genre, width = 25), total_revenue), y = total_revenue)) +
  geom_col(fill = "#796878") +
  geom_text(
    aes(label = paste0(round(total_revenue / 1e6, 1), "M")),
    hjust = -0.1,
    size = 3
  ) +
  coord_flip() +
  scale_y_continuous(
    labels = label_number(scale = 1e-6, suffix = "M"),
    expand = expansion(mult = c(0, 0.1))
  ) +
  labs(
    title = "Total Revenue by Genre",
    x = "Genre",
    y = "Total Revenue (Millions USD)"
  ) +
  theme_sage_purple() +
  guides(fill = "none")

```



2. Which sub-genres generate the most total revenue? (using number of reviews multiplied by final selling price)

```
Dataset <- Dataset |>
  mutate(revenue2 = final_price * reviews_count)
```

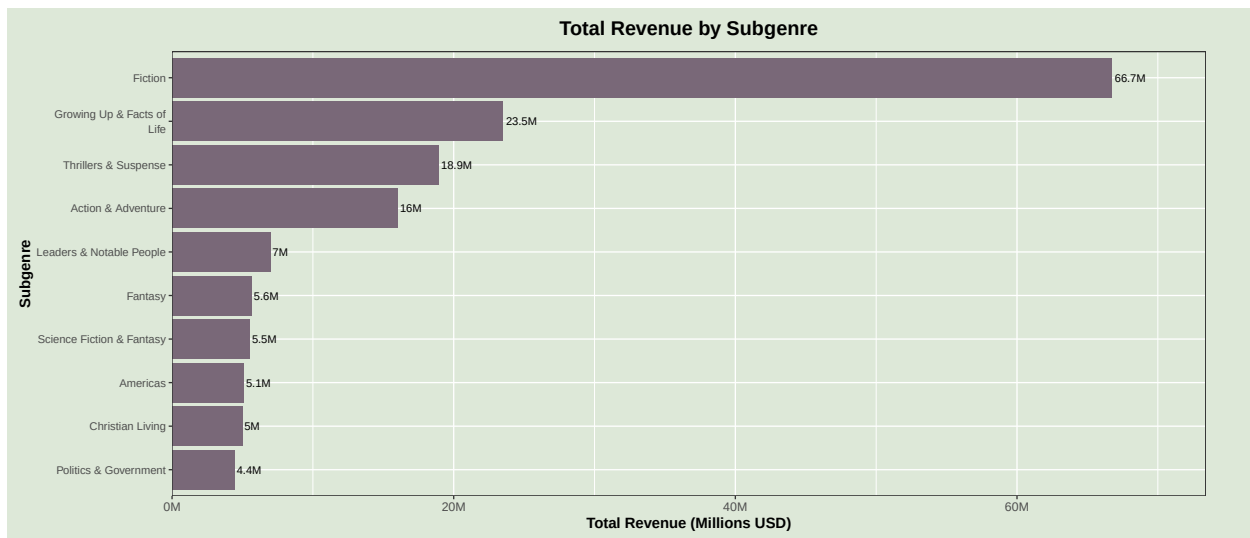
```
subgenre_revenue <- Dataset |>
  group_by(subgenre) |>
  summarize(total_revenue = sum(revenue2, na.rm = TRUE)) |>
  arrange(desc(total_revenue)) |>
  slice_head(n = 10)
```

```
ggplot(subgenre_revenue, aes(x = reorder(str_wrap(subgenre, width = 25), total_revenue), y = total_revenue)) +
  geom_col(fill = "#796878") +
  geom_text(
    aes(label = paste0(round(total_revenue / 1e6, 1), "M")),
    hjust = -0.1,
    size = 3
  ) +
  coord_flip() +
  scale_y_continuous(
    labels = label_number(scale = 1e-6, suffix = "M"),
    expand = expansion(mult = c(0, 0.1))
  )
```

```

) +
labs(
  title = "Total Revenue by Subgenre",
  x = "Subgenre",
  y = "Total Revenue (Millions USD)"
) +
theme_sage_purple() +
guides(fill = "none")

```



*3. Which individual bestselling books drive the most revenue?

```

top_books <- Dataset |>
  mutate(revenue = final_price * reviews_count) |>
  arrange(desc(revenue)) |>
  slice_head(n = 10)

print(top_books |>
  select(title, author, genre, subgenre, revenue))

```

```

##                                     title
## 1                                Harry Potter Hardcover Boxed Set
## 2                                Becoming
## 3                                Where The Crawdads Sing
## 4                                The Alchemist Gift Edition
## 5                                Harry Potter Paperback Box Set
## 6  A Game Of Thrones A Clash Of Kings A Storm Of Swords A Feast For Crows A Dance With Dragons
## 7                                A Promised Land
## 8                                The Midnight Library
## 9                                The Wonky Donkey Book Toy Boxed Set
## 10                               Sapiens
##
##          author                genre
## 1      J. K. Rowling      Children's Books
## 2      Michelle Obama      History

```

```

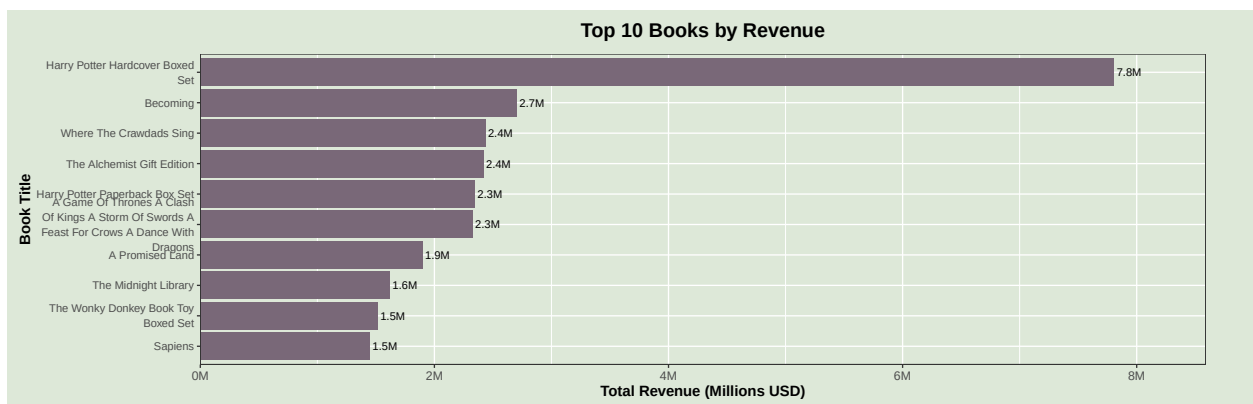
## 3      Delia Owens      Literature & Fiction
## 4      Paulo Coelho      Reference
## 5      J. K. Rowling      Children's Books
## 6      George R. R. Martin      Literature & Fiction
## 7      Barack Obama      Biographies & Memoirs
## 8      Matt Haig      Science Fiction & Fantasy
## 9      Craig Smith      Children's Books
## 10     Yuval Noah Harari      History
##
##      subgenre revenue
## 1      Growing Up & Facts of Life 7808641
## 2      Americas 2706693
## 3      Fiction 2437493
## 4      Writing, Research & Publishing Guides 2420865
## 5      Growing Up & Facts of Life 2348053
## 6      Action & Adventure 2327355
## 7      Leaders & Notable People 1899908
## 8      Science Fiction 1621380
## 9      Growing Up & Facts of Life 1517787
## 10     World 1451210

```

```

ggplot(top_books, aes(x = reorder(str_wrap(title, width = 30), revenue), y = revenue)) +
  geom_col(fill = "#796878") +
  geom_text(
    aes(label = paste0(round(revenue / 1e6, 1), "M")),
    hjust = -0.1,
    size = 3
  ) +
  coord_flip() +
  scale_y_continuous(
    labels = label_number(scale = 1e-6, suffix = "M"),
    expand = expansion(mult = c(0, 0.1))
  ) +
  labs(
    title = "Top 10 Books by Revenue",
    x = "Book Title",
    y = "Total Revenue (Millions USD)"
  ) +
  theme_sage_purple() +
  guides(fill = "none")

```

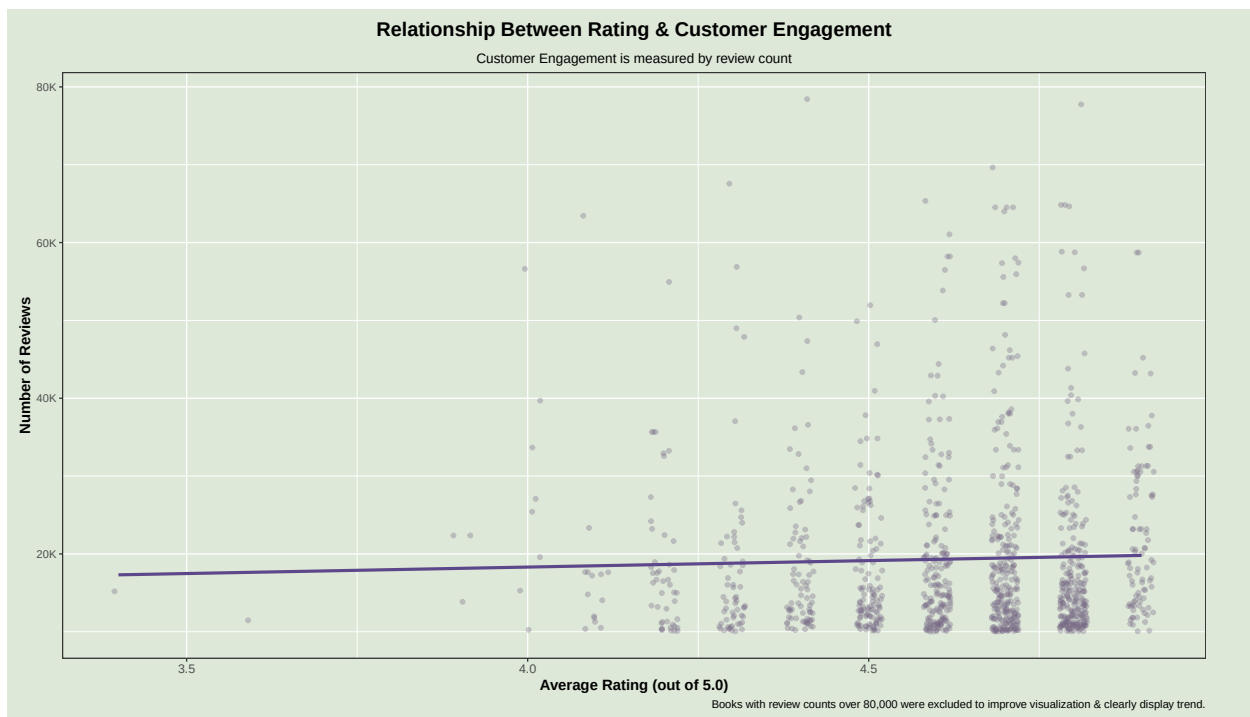


4. Do highly rated books mean stronger customer engagement? (relationship between rating & review)

```
Dataset_filtered <- Dataset |>
  filter(reviews_count < 80000) #remove outlier values to better show trend

ggplot(Dataset_filtered, aes(x = rating, y = reviews_count)) +
  geom_jitter(width = 0.02, alpha = 0.35, size = 1.5, color = "#796B87") +
  geom_smooth(method = "lm", color = "mediumpurple4", se = FALSE, linewidth = 1.2) +
  scale_y_continuous(labels = label_number(scale = 1e-3, suffix = "K")) +
  labs(
    title = "Relationship Between Rating & Customer Engagement",
    subtitle = "Customer Engagement is measured by review count",
    caption = "Books with review counts over 80,000 were excluded to improve visualization & clearly display trend",
    x = "Average Rating (out of 5.0)",
    y = "Number of Reviews"
  ) +
  theme_sage_purple() +
  theme(plot.subtitle = element_text(hjust = 0.5, size = 11))
```

'geom_smooth()' using formula = 'y ~ x'



alternative Boxplot to display info in another way

```
Dataset_filtered <- Dataset |>
  filter(reviews_count < 80000) |>
```

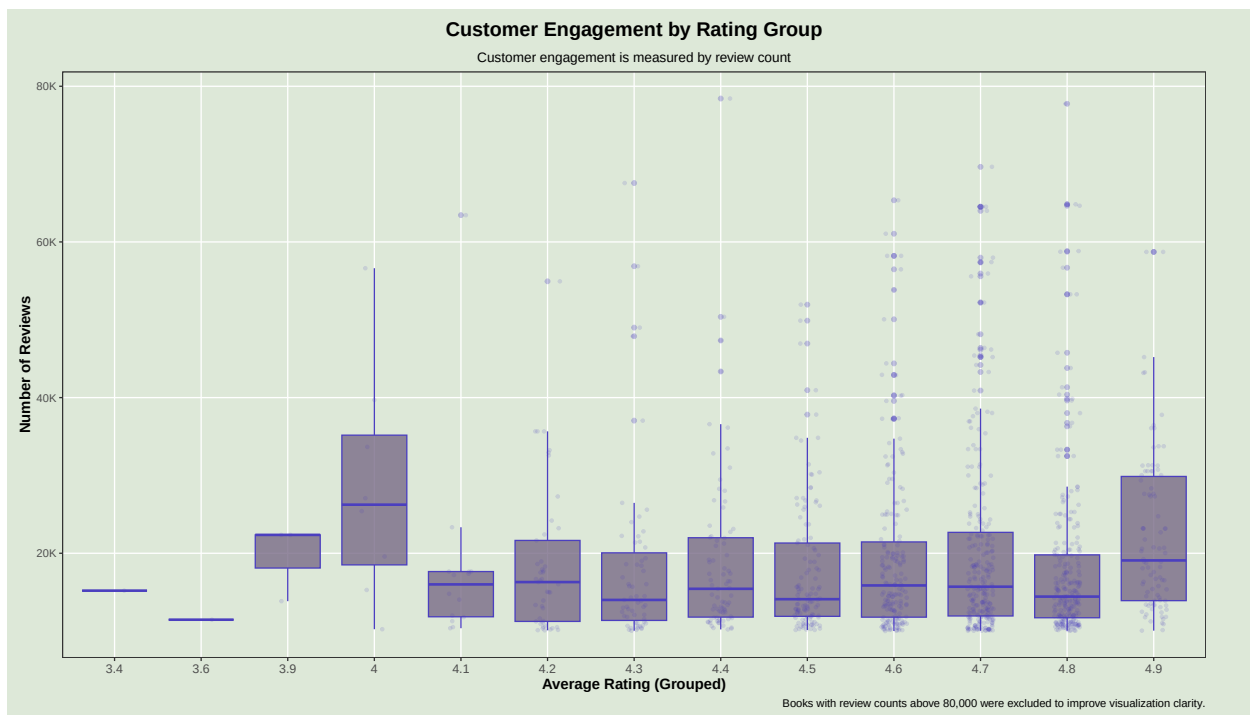


```

mutate(rating_group = factor(round(rating, 1)))

ggplot(Dataset_filtered, aes(x = rating_group, y = reviews_count)) +
  geom_boxplot(fill = "#796B87", color = "#4B3FBF", alpha = 0.8, outlier.alpha = 0.3) +
  geom_jitter(width = 0.15, alpha = 0.15, size = 1, color = "#4B3FBF") +
  scale_y_continuous(labels = label_number(scale = 1e-3, suffix = "K")) +
  labs(
    title = "Customer Engagement by Rating Group",
    subtitle = "Customer engagement is measured by review count",
    x = "Average Rating (Grouped)",
    y = "Number of Reviews",
    caption = "Books with review counts above 80,000 were excluded to improve visualization clarity."
  ) +
  theme_sage_purple() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 16, face = "bold"),
    plot.subtitle = element_text(hjust = 0.5, size = 11),
    panel.grid.minor = element_blank()
  )

```



5. Which books should be prioritized moving forward? (revenue & engagement)

```

books_priority <- Dataset |>
  mutate(
    revenue = final_price * reviews_count,
    revenue_z = as.numeric(scale(revenue)),
    engagement_z = as.numeric(scale(reviews_count)),
    priority_score = 0.6 * revenue_z + 0.4 * engagement_z
  )

```

```

) |>
  arrange(desc(priority_score))

head(books_priority |>
  select(title, author, genre, subgenre, revenue, reviews_count, priority_score), 10)

```

```

##                                     title
## 1                                Harry Potter Hardcover Boxed Set
## 2                                Where The Crawdads Sing
## 3                                Becoming
## 4                                A Promised Land
## 5                                The Midnight Library
## 6                                The Alchemist Gift Edition
## 7                                Harry Potter Paperback Box Set
## 8                                Too Much And Never Enough
## 9 A Game Of Thrones A Clash Of Kings A Storm Of Swords A Feast For Crows A Dance With Dragons
## 10                               The Silent Patient
##
##               author               genre
## 1           J. K. Rowling       Children's Books
## 2           Delia Owens        Literature & Fiction
## 3           Michelle Obama      History
## 4           Barack Obama       Biographies & Memoirs
## 5           Matt Haig          Science Fiction & Fantasy
## 6           Paulo Coelho       Reference
## 7           J. K. Rowling       Children's Books
## 8 by Mary L. Trump Ph.D. (Author) Biographies & Memoirs
## 9           George R. R. Martin Literature & Fiction
## 10          Alex Michaelides Mystery, Thriller & Suspense
##
##               subgenre revenue reviews_count priority_score
## 1           Growing Up & Facts of Life 7808641      58716      13.291976
## 2           Fiction 2437493      196572      8.445390
## 3           Americas 2706693      117478      6.664023
## 4           Leaders & Notable People 1899908      118596      5.386698
## 5           Science Fiction 1621380      122000      5.030359
## 6 Writing, Research & Publishing Guides 2420865      64522      4.715340
## 7           Growing Up & Facts of Life 2348053      58716      4.434414
## 8           Leaders & Notable People 1296774      100137      3.890715
## 9           Action & Adventure 2327355      36948      3.790399
## 10          Thrillers & Suspense 1202056      92537      3.523946

```

```

top_priority <- books_priority |>
  slice_head(n = 10)

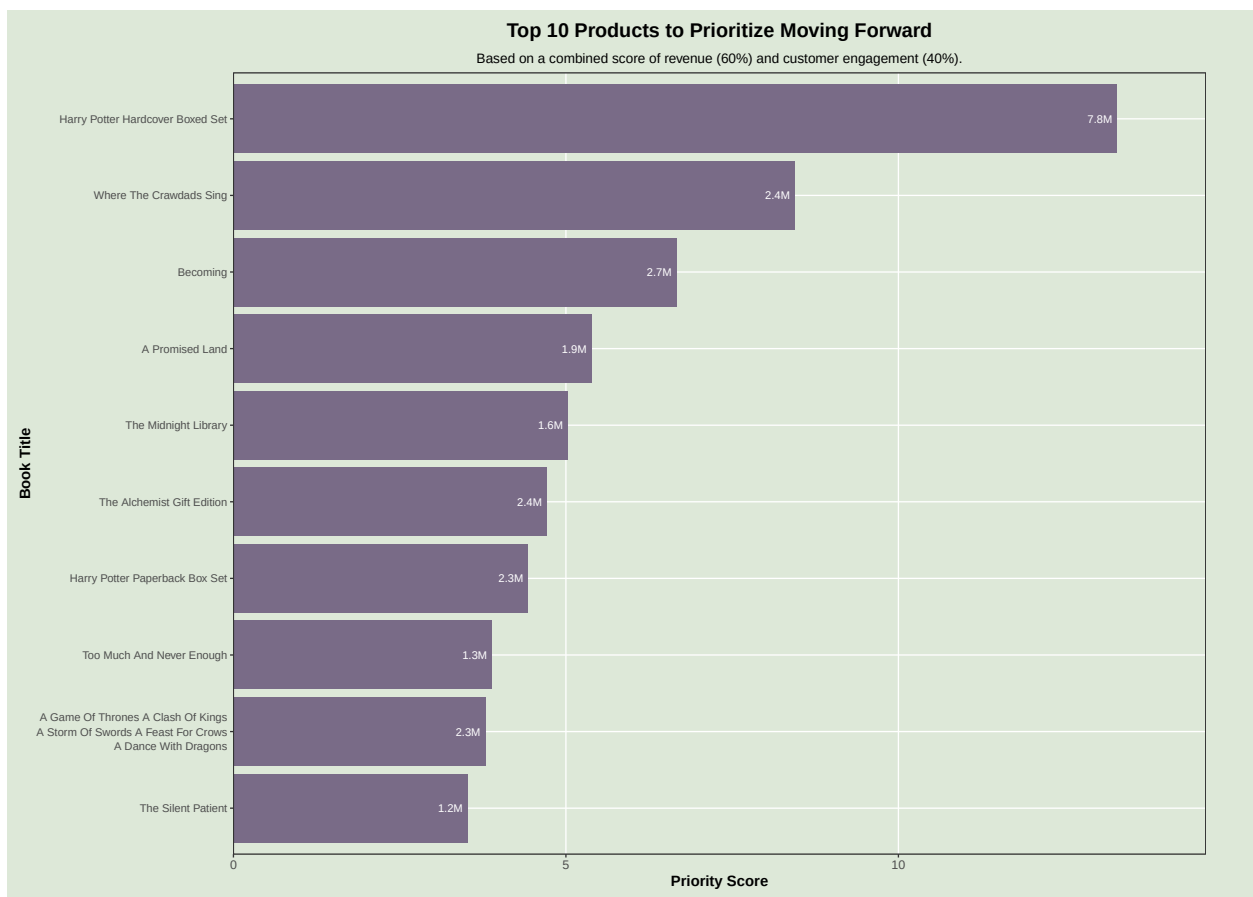
ggplot(top_priority, aes(x = reorder(str_wrap(title, width = 35), priority_score), y = priority_score))
  geom_col(fill = "#796B87") +
  geom_text(
    aes(label = paste0(round(revenue / 1e6, 1), "M")),
    hjust = 1.2,
    color = "white",
    size = 3
  ) +
  coord_flip() +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1)))

```

```

) +
labs(
  title = "Top 10 Products to Prioritize Moving Forward",
  subtitle = "Based on a combined score of revenue (60%) and customer engagement (40%).",
  x = "Book Title",
  y = "Priority Score"
) +
theme_sage_purple() +
theme(
  plot.title = element_text(hjust = 0.5, size = 16, face = "bold"),
  plot.subtitle = element_text(hjust = 0.5, size = 11),
  panel.grid.minor = element_blank()
) +
guides(fill = "none")

```



Experimental to show contributions to Priority Score

```

top_priority_long <- top_priority |>
  select(title, priority_score, revenue_z, engagement_z) |>
  mutate(title = str_wrap(title, width = 30))

p1 <- top_priority_long |>

```

```

select(title, priority_score) |>
ggplot(aes(x = reorder(title, priority_score), y = priority_score)) +
geom_col(fill = "#796B87") +
geom_text(aes(label = round(priority_score, 2)), hjust = -0.1, size = 3) +
coord_flip() +
scale_y_continuous(expand = expansion(mult = c(0, 0.1))) +
labs(
  title = "Priority Score Ranking",
  x = "Book Title",
  y = "Priority Score"
) +
theme_sage_purple() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold"),
  panel.grid.minor = element_blank()
)

p2 <- top_priority_long |>
pivot_longer(
  cols = c(revenue_z, engagement_z),
  names_to = "component",
  values_to = "value"
) |>
mutate(
  component = recode(component,
    revenue_z = "Revenue",
    engagement_z = "Engagement")
) |>
ggplot(aes(x = reorder(title, value), y = value, fill = component)) +
geom_col(position = "dodge") +
coord_flip() +
scale_fill_manual(values = c("Revenue" = "#796B87", "Engagement" = "#4B3FBF")) +
labs(
  title = "Revenue vs Engagement Contribution",
  x = "Book Title",
  y = "Standardized Contribution",
  fill = "Component"
) +
theme_sage_purple() +
theme(
  plot.title = element_text(hjust = 0.5, face = "bold"),
  panel.grid.minor = element_blank()
)

p2

```

